

3.1 The /proc File System

3.1.1 Virtually speaking

Having come this far with Linux, have you wondered if there is a way in which an advanced user (you can be one too, with some practice and

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kailas@kailas:~$ ls /proc
1      5526 6350 6495 6568 6845  cpuinfo  ioports  partitions
112    5528 6363 6497 6571 6872  crypto   irq      ramfs
113    5541 6376 6501 6582 6877  devices  kallsyms slabinfo
114    5554 6412 6503 6586 6878  diskstats kcore    stat
115    5559 6413 6506 6588 6890  dma      kmsg     swaps
1593   5657 6414 6519 6590 7      dri      loadavg  sys
2      5662 6415 6537 6622 700    driver   locks    sysrq-trigger
2452   6011 6416 6548 6626 84     execdomains mdstat  sysvipc
2578   6038 6417 6550 6639 acpi     fb        meminfo  tty
3      6063 6448 6552 6642 asound   filesystems misc      uptime
4      6119 6486 6556 6643 bluetooth fs        modules  version
5      6313 6489 6560 6644 buddyinfo ide        mounts   vmstat
5494   6321 6490 6562 6786 bus       interrupts mtrr
5509   6331 6492 6564 6799 cmdline  iomem    net
kailas@kailas:~$
```

A typical /proc file system contents

patience) will be able to take a peek at the kernel and understand what is happening at any given moment? Well, Linux gives a user more than enough information in the form of the “/proc” file system.

The /proc file system is a “virtual” file system that does not reside on any physical media. It is a file system that resides in the memory and contains real-time information about the devices that make up the PC, the processes that are running and the processes that are in a sleep state. Some of this information is readable and comprehensible, while some is not. The information needs to be interpreted, and to do this you have many utilities at your disposal.

Before we proceed, let’s give you a small sample of the kind of things a user can easily access. The GUI tools that can give you system information will also read from the /proc. Let us say you want to view information about your CPU. Just get into the terminal (console) and type in “cat /proc/cpuinfo”.

This will give you information about your CPU. But that’s not all. You can also view running processes and what resources they are consuming, the kernel, etc. While a new user may be confused by the sheer amount of information that /proc offers, a developer or enthusiast will find a lot of useful information here. Most infor-